



ANNE GRACIA A

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PROFESSIONAL SUMMARY

Chemical Engineer and Materials Researcher working at the intersection of experimental science, artificial intelligence, and industrial innovation. Experienced in translating materials research into practical engineering solutions through funded research, industrial R&D, and AI-assisted materials discovery. My work spans corrosion science, surface engineering, thin-film deposition, scientific literature intelligence, uncertainty quantification, and data-driven research workflows. Particularly interested in how interdisciplinary innovation can accelerate sustainable technologies and create meaningful real-world impact.

WORK EXPERIENCE

N-ERGY AI SOLUTIONS INDIA PVT LTD

AI Materials associate | Oct 2025 - Present

- Contribute to the development of AI-driven materials discovery workflows designed to accelerate scientific research and materials R&D.
- Part of the core product development team, contributing to the product vision, workflow design, and development of the platform's core research capabilities.
- Led the development and implementation of uncertainty quantification workflows integrated into AI-assisted materials research, enabling more reliable assessment of model predictions and research outcomes.

Experimental Materials associate | Feb 2025 - Oct 2025

- Supported scientists in experimental planning, execution, and materials evaluation while contributing to the development of materials solutions for industrial requirements.
- Led experimental investigation and Optimized Aluminum thin-film deposition using RF sputtering to achieve uniform 100 nm films for micro-device architectures, and identifying a viable solution within four weeks.

RESEARCH & INNOVATION PROJECTS

Undergraduate Researcher - Institutionally Funded Project | SSN College, Chennai, India | Jan 2024 – Oct 2024

Guide: Dr. A.K.Lakshminarayanan

- Investigated the electrochemical kinetics and pitting corrosion behavior of WAAM 316L stainless steel using advanced potentiodynamic polarization and microscopy
- Received ₹35,000 in institutional research funding supporting experimental design and materials characterization.

Project Intern - Industrial Surface Engineering | BRAKES INDIA PVT LTD, Chennai, India | Oct 2024 – Feb 2025

Guide: Dr. Pachimatla Rajesh

- Optimized industrial electrochemical anodizing processes for automotive components, achieving a reported 78% improvement in surface microfinish.
- Presented findings at the National Conference on Innovation of Process Engineering for Sustainable Development, IChE-CRC, March 2025.

PUBLICATIONS/CONFERENCES:

National Conference on Innovation of Process Engineering for Sustainable Development

Presented research at the National Conference on Innovation of Process Engineering for its Sustainable Development, IICChE-CRC | March 2025.

Paper: "*Enhancing Microfinish and Resolving Smut and Powdery Coating Issues in the Anodizing Process for Automotive Plunger Manufacturing*"

ACHIEVEMENTS, LEADERSHIP & COMMUNITY ENGAGEMENT

- Department 3rd Rank, B.Tech Chemical Engineering, SSN College of Engineering.
- Recipient of ₹35,000 Institutional Research Funding for undergraduate research on the electrochemical kinetics and pitting corrosion behaviour of WAAM 316L stainless steel.
- District-Level Handball Player, representing at the district level in competitive handball.
- Department Event Organizer, contributed to the organization and execution of a departmental technical symposium INVENTE 2023 - 2024.
- Youth Red Cross Volunteer, participated in blood donation drives, food distribution campaigns, community outreach camps, and social-service initiatives.
- Community Education Volunteer, supported NGO-led teaching initiatives for underprivileged children.
- Community Service Volunteer, participated in government-school repainting and community improvement initiatives.

RESEARCH INTERESTS

- Sustainable Materials & Manufacturing
- AI for Scientific Discovery
- Materials Innovation & Advanced Manufacturing
- Industrial R&D and Process Optimization
- Energy & Functional Materials
- Technology for Sustainable Development

TECHNICAL SKILLS

- Materials Characterization: XRD, XPS, SEM/EDS, AFM, FTIR; structural, chemical, and surface analysis
- Materials Processing & Synthesis: RF sputtering, sol-gel synthesis, solid-state metallurgy, arc melting, high-entropy alloys, CO₂ laser processing
- Data Analysis & Scientific Computing: Python (Pandas, Plotly), MATLAB, OriginLab, Aspen Plus, Streamlit
- Research & Development: Uncertainty Quantification, scientific literature analysis, data visualization, AI-assisted research workflows
- Development Tools: Git/GitHub, PyMongo, Streamlit deployment

ACADEMIC HISTORY

SSN College of Engineering | 2021-2025

Bachelor of Technology in Chemical Engineering

- CGPA: 8.3/10 (equivalent to ~3.3/4 GPA)
- Department 3rd Rank

SBOA School and Junior College | 2019-2021

High school

- Percentage: 88%